Year 2 Pure Mathematics

Chapter 3

Sequences and series

Practice Booklet

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3.1 Arithmetic sequences

Practice Book, printed page 17.

1. For each arithmetic sequence:

i. write down the first 4 terms of the sequence

ii. write down the values of a and d.

a. $4n + 3$

b. $7 - 3n$

c. $8 + \frac{1}{2}n$

d. $2n - 5$

Hint

In an **arithmetic sequence**, the difference between consecutive terms is constant.
The formula for the nth term, u_n, of an arithmetic sequence is $u_n = a + (n-1)d$, where a is the first term of the sequence and d is the common difference.

2. For each arithmetic sequence, find an expression for the nth term and find the 10th term.

a. $2, 6, 10, 14, \ldots$

b. $5, 3, 1, -1, \ldots$

c. $-2, 1, 4, 7, \ldots$

d. $8y, 5y, 2y, -y, \ldots$

3. Calculate the number of terms in each arithmetic sequence.

a. $2, 5, 8, \ldots, 95, 98$

b. $75, 73, 71, \ldots, -3, -5$

c. $-3, 1, 5, \ldots, 109, 113$

d. $-4k, -2k, 0, \ldots, 98k, 100k$

Hint

Find an expression for the nth term, u_n, and set it equal to the final term in the sequence.

4. In an arithmetic sequence, the fourth term is 17 and the 10th term is 47.
Find the first term and the common difference.

Hint

Use the general form for the nth term to set up two simultaneous equations in a and d.
← Year 1, Section 3.1

5. The first term of an arithmetic sequence is 20 and the common difference is $-\frac{1}{2}$

a. Find the value of the 25th term.

The rth term of the sequence is 0.

b. Find the value of r.

6. The nth term of an arithmetic sequence is $3n - 4$.

a. Write down the first three terms of the sequence.

b. State the value of the common difference.

7. E/P An arithmetic sequence has first term a and common difference d.
The fourth term of the sequence is -2 and the ninth term is -22.

a. Find the value of a and the value of d. (3 marks)

The nth term of the sequence is -54.

b. Find the value of n. (3 marks)

8. E/P a, b and c are three consecutive terms in an arithmetic sequence.

a. Show that $\frac{a+c}{2} = b$. (3 marks)

The first three terms of a different arithmetic sequence are 15, p^2 and $-p$, where p is a constant.

b. Find the two possible values of p. (3 marks)

9. E/P The first three terms of an arithmetic sequence are k, $4k - 3$ and $3k + 10$.

a. Find the value of k. (3 marks)

b. Find the value of the 50th term of this sequence. (3 marks)

3.2 Arithmetic series

Practice Book, printed page 18.

1. Find the sums of the following arithmetic series:

a. $5 + 9 + 13 + 17 + \ldots$ (20 terms)

b. $3 + 8 + 13 + 18 + \ldots$ (15 terms)

c. $12 + 10 + 8 + 6 + \ldots$ (40 terms)

d. $8 + 7.5 + 7 + 6.5 + \ldots$ (25 terms)

Hint

$S_n = \frac{n}{2}(2a + (n-1)d)$

2. Find the sums of the following arithmetic series:

a. $6 + 9 + 12 + \ldots + 51$

b. $52 + 47 + 42 + \ldots + (-103)$

c. $-2 + 1 + 4 + \ldots + 175$

d. $120 + 112 + 104 + \ldots + 0$

Hint

Start by working out the value of n from the given last term.

3. Prove that the sum of the first 60 natural numbers is 1830.

Hint

The natural numbers are the positive integers, $1, 2, 3, 4, \ldots$.

$$S_{60} = 1 + 2 + 3 + \ldots + 59 + 60$$

$$S_{60} = 60 + 59 + 58 + \ldots + 2 + 1$$

Work out $2S_{60}$ and divide the result by 2.

4. Find the least number of terms required for the sum of the arithmetic series $5 + 8 + 11 + 14 + \ldots$ to exceed 1000.

Hint

Find the values of a and d for this series, then find the least positive integer n that satisfies

$$S_n = \frac{n}{2}(2a + (n-1)d) > 1000$$

5. The rth term of an arithmetic series is $4r - 5$.

a. Write down the first three terms of the series.

b. State the value of the common difference.

c. Show that $S_n = n(2n - 3)$.

6. E/P

a. Prove that the sum of the first n natural numbers is $\frac{n(n+1)}{2}$.

You may not use the formula for the sum of an arithmetic series in your proof. (3 marks)

b. Hence find the sum of the even integers from 2 to 200. (2 marks)

7. P $u_1, u_2, \ldots, u_n$ are the terms of an arithmetic sequence.

a. Given $u_4 = 103$ and $u_{12} = 79$, find the value of u_{25}.

The sum of the first n terms of the sequence is 1925.

b. Find the value of n.

8. E An arithmetic sequence has first term a and common difference d.

The sum of the first 12 terms of the sequence is 366.

a. Show that $12a + 66d = 366$. (2 marks)

Given also that the eighth term of the sequence is 38,

b. write down a second equation in a and d (1 mark)

c. find the value of a and the value of d. (4 marks)

9. E/P An arithmetic sequence has first term a and common difference d.

The sum of the second term and the sixth term is 14.

a. Show that $a + 3d = 7$. (3 marks)

The sum of the first 20 terms of the sequence is -120.

b. Find the value of a and the value of d. (4 marks)

10. E/P The first term of an arithmetic series is a and the common difference is d.

The 15th term of the series is 11 and the 20th term of the series is $\frac{37}{2}$

a. Show that that the first term of the series is -10, and write down the second term. (4 marks)

The sum of the first n terms of the series is 2055.

b. Find the value of n. (4 marks)

11. E/P The second, third and fourth terms of an arithmetic sequence are $3k$, $4k + 5$ and $7k$ respectively, where k is a constant. Show that the sum of the first n terms of the sequence is kn^2, where k is a rational number to be found. (5 marks)

12. E/P The third, fourth and fifth terms of an arithmetic sequence are $3k$, $4k + 3$ and $6k - 9$ respectively, where k is a constant. Show that the sum of the first n terms of the sequence is a square number. (5 marks)

3.3 Geometric sequences

Practice Book, printed page 20.

1. For each geometric sequence, find:

i. the common ratio

ii. the nth term.

a. $2, 6, 18, 54, \ldots$

b. $60, 30, 15, 7.5, \ldots$

c. $10, -20, 40, -80, \ldots$

d. $40, 60, 90, 135, \ldots$

Hint

The formula for the nth term, u_n, of a geometric sequence is $u_n = ar^{n-1}$, where a is the first term and r is the common ratio.

2.

a. Write down the sequence from question 1 which is alternating.

b. Given that p, q and r are three consecutive terms in a geometric sequence, show that $pr = q^2$.

Hint

A geometric sequence alternates if the terms are alternately positive and negative.

3. The terms 5, x and $x + 10$ form the first three terms of a geometric sequence.
Given that the terms of the sequence are positive, find:

a. the value of x

b. the 10th term of the sequence.

Hint

$$\frac{u_2}{u_1} = \frac{u_3}{u_2}$$

4. Find the first term in the geometric progression $4, 12, 36, 108, \ldots$ that is larger than 1 000 000.

Hint

$a = 4$ and $r = 3$, so you need to find the smallest positive integer that satisfies

$$4 \times 3^{n-1} > 1\,000\,000$$

You can solve this inequality by taking logs on both sides. ← Year 1, Section 14.6

5. P The fourth term of a geometric series is 16 and the seventh term of the series is 250.
For this series, find:

a. the common ratio

b. the first term.

6. E/P The first three terms of a geometric sequence are $k+4$, $3k$ and $2k^2$, where k is a positive constant. Find the value of k. (4 marks)

7. E/P The third term of a geometric sequence is 108 and the sixth term is 32. Find:

a. the 20th term (3 marks)

b. the first term in the sequence which is less than 1. (3 marks)

8. E/P A geometric sequence has first term 150 and common ratio r, where $r > 0$.
The fifth term of the sequence is 50.

a. Show that r satisfies $4 \ln r + \ln 3 = 0$. (3 marks)

b. Hence, or otherwise, find the value of r correct to 3 significant figures. (1 mark)

9. E/P The first three terms of a geometric series are $5k$, $4k + 3$ and $6k + 7$, where k is a positive constant. Find:

a. the value of k (6 marks)

b. the first term in the sequence to exceed 10 000. (3 marks)

3.4 Geometric series

Practice Book, printed page 21.

1. Find the sums of the following geometric series. Give your answers to 3 significant figures if necessary.

a. $2 + 6 + 18 + 54 + \ldots$ (10 terms)

b. $48 + 24 + 12 + 6 + \ldots$ (12 terms)

c. $3 - 6 + 12 - 24 + \ldots$ (8 terms)

d. $810 + 540 + 360 + 240 + \ldots$ (9 terms)

Hint

The sum of the first n terms of a geometric series is given by the formula

$$S_n = \frac{a(r^n - 1)}{r - 1}, \ r \neq 1 \qquad \text{(easier to use if } |r| > 1\text{)}$$

or $S_n = \dfrac{a(1 - r^n)}{1 - r}$, $r \neq 1$ (easier to use if $|r| < 1$)

2. Find the sums of the following geometric series. Give your answers to 3 significant figures if necessary.

a. $5 + 10 + 20 + \ldots + 640$

b. $13\,122 + 4374 + 1458 + \ldots + 2$

c. $1024 - 256 + 64 - \ldots - \dfrac{1}{16}$

d. $128 + 192 + 288 + \ldots + 1458$

Hint

First find the number of terms, n, using nth term $= ar^{n-1}$.

3. Find the least value of n such that the sum of the first n terms of the geometric series $3+6+12+24+\ldots$ exceeds 1 000 000.

Hint

Work out the values of a and r from the terms given. Use the formula for the sum to n terms to set up an inequality, and use logs to find the least positive integer that satisfies the inequality.

4. P The fourth term of a geometric series is 24. The fifth term of the series is 48.
For this series, find:

a. the common ratio

b. the first term

c. the sum of the first 20 terms.

5. E A geometric series has first term 10 and common ratio $\dfrac{3}{5}$. Calculate:

a. the 10th term of the series to 3 decimal places (2 marks)

b. the least value of n such that the sum to n terms is greater than 20. (4 marks)

6. E/P A geometric series has first term 20 and common ratio $\frac{4}{5}$

Given that the sum to k terms is greater than 50,

a. show that $k > \frac{\ln 0.5}{\ln 0.8}$ (4 marks)

b. find the smallest possible value of k. (1 mark)

7. E/P The first three terms of a geometric series are $2k - 4$, $k + 2$ and $k - 1$, where k is a positive integer.

a. Find the value of k. (6 marks)

b. Hence find the 20th term of the series, correct to 3 significant figures. (2 marks)

c. Find the sum of the first 20 terms of this series, correct to 3 significant figures. (3 marks)

8. E/P The second term of a geometric series is 96. Given that $a + ar + ar^2 = 292.8$, where a is the first term of the series and $r < 1$ is the common ratio, find the sum of the first 10 terms of the series. Give your answer to 3 significant figures. (9 marks)

9. E/P A geometric series is such that the sum of the first two terms is equal to the third term.

a. Show that the common ratio of the series is independent of the first term. (2 marks)

Given further that the common ratio is positive and that the first term of the series is 2,

b. find the sum of the first 20 terms of the series, correct to 3 significant figures. (8 marks)

3.5 Sum to infinity

Practice Book, printed page 22.

1.

a. Explain why the geometric series $2 + 0.2 + 0.02 + 0.002 + \ldots$ is convergent.

b. Find the sum to infinity of this series.

Hint

A geometric series is convergent if $|r| < 1$ (or $-1 < r < 1$).
This means that the sum of the first n terms in the series tends to a limit as n increases.
This limit is called the sum to infinity of the series and can be found using the formula $S_\infty = \frac{a}{1-r}$.

2. For each of the following series:

i. state, with a reason, whether the series is convergent.

ii. If the series is convergent, find the sum to infinity.

a. $5 + 2 + 0.8 + 0.32 + \ldots$

b. $0.1 + 0.3 + 0.9 + 1.2 + \ldots$

c. $81 - 27 + 9 - 3 + \ldots$

d. $20 + 18 + 16 + 14 + \ldots$

3. The fifth term of a geometric series is 2.4576 and the seventh term is 1.572864.

a. Explain why this series is convergent.

b. Find the sum to infinity of the series.

Hint

You can use the general form of the nth term of a geometric series to find the values of a and r.

4. A geometric series has first term 8 and sum to infinity 25.

a. Find the common ratio.

A different geometric series has sum to infinity 40 and common ratio $\frac{1}{5}$

b. Find the first term.

Hint

Substitute into $S_\infty = \dfrac{a}{1-r}$

5. For a geometric series with first term a and common ratio r, $S_4 = 80$ and $S_\infty = 81$.

a. Find the possible values of r.

b. Given that all the terms in the series are positive, find the value of a.

Hint

$S_n = \dfrac{a(1-r^n)}{1-r}$ and $S_\infty = \dfrac{a}{1-r}$,
so $S_n = S_\infty \times (1-r^n)$

6. E The first term of a geometric progression is 4 and the sum to infinity is 12.
Find the common ratio. (3 marks)

7. E

a. State the condition for an infinite series with common ratio r to be convergent. (1 mark)

b. Find the sum to infinity of the geometric series $\dfrac{3}{8} + \dfrac{3}{16} + \dfrac{3}{32} + \ldots$ (3 marks)

8. E/P In a convergent geometric series, the common ratio is r and the first term is 4.
Given that $S_\infty = 20 \times S_4$, find:

a. the value of the common ratio, giving your answer to 4 significant figures (3 marks)

b. the value of the fifth term. (2 marks)

9. E/P The first term of a geometric series is 80. The sum to infinity is 120.

a. Find the common ratio. (3 marks)

b. Find, to 2 decimal places, the difference between the fourth and fifth terms. (2 marks)

c. Calculate the sum of the first 6 terms. (2 marks)

10. E/P A geometric series has first term a and common ratio r.
The second term of the series is $\dfrac{12}{5}$ and the sum to infinity of the series is 10.

a. Show that $50r^2 - 50r + 12 = 0$. (4 marks)

b. Find the two possible values of r. (2 marks)

c. Find the corresponding two possible values of a. (2 marks)

Given that r takes the smaller of its two possible values,

d. find the smallest value of n for which S_n exceeds 9.99. (2 marks)

3.6 Sigma notation

Practice Book, printed page 23.

1. For each series:

i. write out every term in the series.

ii. Hence find the value of the sum.

a. $\displaystyle\sum_{r=1}^{6}(4r-3)$

b. $\displaystyle\sum_{r=1}^{5} 2r^3$

c. $\displaystyle\sum_{r=0}^{4} \cos(90r^\circ)$

d. $\sum_{r=3}^{7} 3\left(-\frac{1}{2}\right)^r$

Hint

The Greek capital letter 'sigma' is used to represent a sum. For example,

$$\sum_{r=1}^{4}(2r+1) = 3+5+7+9, \text{ which is an arithmetic}$$

series with $a = 3$, $d = 2$ and $n = 4$.

2. An arithmetic sequence has nth term $u_n = 5n - 1$.

a. Write down the first term and the common difference of the sequence.

b. Hence find $\sum_{r=1}^{12}(5r-1)$.

3. A geometric sequence has nth term

$$u_n = 3 \times 2^{n-1}$$

a. Write down the first term and the common ratio of the sequence.

b. Hence find $\sum_{r=1}^{20} 3 \times 2^{r-1}$

Hint

$\sum_{r=1}^{n} a \times b^{r-1} = a + ab + ab^2 + ab^3 + \ldots + ab^{n-1}$
is the sum of an n term geometric sequence with first term a and common ratio b.

4. Find:

a. $\sum_{r=1}^{10}(3r+2)$

b. $\sum_{r=1}^{20} 3$

c. $\sum_{r=1}^{\infty} 5\left(\frac{2}{3}\right)^r$

Hint

In part b there is no r in the expression following the summation, so 3 is added to itself 20 times.
Part c is a geometric series. Find the first term and the common ratio.

5. $\sum_{r=8}^{30}(5-2r)$ represents an arithmetic series.

a. Explain why

$$\sum_{r=8}^{30}(5-2r) = \sum_{r=1}^{30}(5-2r) - \sum_{r=1}^{7}(5-2r)$$

b. Hence find $\sum_{r=8}^{30}(5-2r)$.

Hint

Use $\sum_{r=1}^{n}(5-2r) = \sum_{r=1}^{m}(5-2r) + \sum_{r=m+1}^{n}(5-2r)$

6. E Find the value of $\sum_{r=10}^{15} 4 \times 3^{r-1}$ (3 marks)

7. E Find $\sum_{r=1}^{10} 10 \times 3^r$ (3 marks)

8. E/P Given that $\sum_{r=1}^{k}(3r+6) = 750$,

a. show that $(k-20)(3k+75) = 0$. (3 marks)

b. Hence find the value of k. (1 mark)

9. E/P Given that $\sum_{r=1}^{k} 4 \times 2^r = 262\,136$,

a. show that $k = \dfrac{\ln 32\,768}{\ln 2}$ (4 marks)

b. For this value of k, calculate $\sum_{r=k+1}^{20} 4 \times 2^r$. (3 marks)

10. E/P The infinite geometric series given by $1 + 4x + 16x^2 + \ldots$ is convergent.

a. Write down the range of possible values of x. (3 marks)

Given that $\sum_{r=1}^{\infty}(4x)^{r-1} = 5$,

b. find the value of x. (3 marks)

11. E/P Show that $\sum_{r=1}^{10}(4 + 3r + 2^{r-1}) = 1228$. (4 marks)

3.7 Recurrence relations

Practice Book, printed page 25.

1. Find the first four terms of the sequences defined by the following recurrence relations:

a. $u_{n+1} = 2u_n - 1$, $u_1 = 3$

b. $u_{n+1} = u_n - 4$, $u_1 = 10$

c. $u_{n+1} = \dfrac{(u_n)^2}{2}$, $u_1 = 4$

Hint

A recurrence relation defines each term in a sequence in terms of the previous term.

2. For each sequence:

i. state whether the sequence is increasing, decreasing or periodic

ii. if the sequence is periodic, write down its order.

a. $3, 7, 11, 15, 19, \ldots$

b. $1, 0, -1, 0, 1, 0, -1, 0, 1, \ldots$

c. $11, 8, 5, 2, -1, \ldots$

Hint

In an **increasing** sequence $u_{n+1} > u_n$ for all $n \in \mathbb{N}$.
In a **decreasing** sequence $u_{n+1} < u_n$ for all $n \in \mathbb{N}$.
In a **periodic** sequence, the terms repeat themselves in a cycle. This means there is an integer k such that $u_{n+k} = u_n$ for all $n \in \mathbb{N}$. The value k is called the order of the sequence.

3. For each sequence:

i. write down the first 5 terms of the sequence

ii. state whether the sequence is increasing, decreasing or periodic

iii. if the sequence is periodic, write down its order.

a. $u_n = 3n - 1$

b. $u_n = 2^{1-n}$

c. $u_n = \sin(180n - 30°)$

d. $u_{n+1} = 2u_n + 1$, $u_1 = 1$

e. $u_{n+1} = 4 - u_n$, $u_1 = 8$

f. $u_{n+1} = u_n - 4$, $u_1 = 10$

4. A sequence $a_1, a_2, a_3, \ldots$ is defined by

$$a_1 = k$$

$$a_{n+1} = a_n + (-1)^n$$

a. Show that $a_2 = a_4$.

Given that $k = 3$,

b. find $\sum_{r=1}^{400} a_r$

c. write down the value of a_{399}.

Hint

$a_2 = k + (-1)^2 = k + 1$
$a_3 = (k + 1) + (-1)^3 = k$, and so on.
The sequence is periodic. You can find the sum of the values in one period and use this to find the sum of the first 400 terms.

5. E/P A sequence is defined for $n \geq 1$ by the recurrence relation $u_{n+1} = pu_n + q$, $u_1 = 4$ where p and q are constants. Given that $u_2 = 3$ and $u_3 = 1$, find the values of p and q. (4 marks)

6. E/P A sequence is given by $u_1 = 5$, $u_{n+1} = ku_n - 8$, where k is an integer.

a. Show that $u_3 = 5k^2 - 8k - 8$. (2 marks)

b. Given that $u_3 = 40$, find the value of k. (3 marks)

c. Hence find the value of u_5. (2 marks)

7. E/P A sequence has nth term $u_n = \sin(90n°)$, $n \geq 1$.

a. Find the order of the sequence. (1 mark)

b. Find $\sum_{r=1}^{200} u_r$ (2 marks)

8. E/P A sequence of numbers $a_1, a_2, a_3, \ldots$ is defined by $a_1 = k$, $a_{n+1} = 5a_n + 4$, $n \in \mathbb{N}$.

a. Find a_3 in terms of k. (2 marks)

b. Show that $\sum_{r=1}^{4} a_r$ is a multiple of 4. (3 marks)

9. E/P A sequence of numbers $a_1, a_2, a_3, \ldots$ is defined by $a_1 = 3$, $a_{n+1} = 1 - \dfrac{1}{a_n - 2}$, $n \in \mathbb{N}$.

a. Find $\sum_{r=1}^{50} a_r$ (3 marks)

b. Hence find $\sum_{r=1}^{50} a_r + \sum_{r=1}^{49} a_r$ (1 mark)

3.8 Modelling with series

Practice Book, printed page 26.

1. Matt is training to compete in a marathon. He starts with a 6 km run one Sunday.
He then increases his run by 2 km each following Sunday.

a. Show that on the fifth Sunday of training he runs 14 km.

b. Find an expression, in terms of n, for the length of his run on the nth Sunday.

A marathon is 42 km.

c. After how many weeks will Matt be running the length of a full marathon?

Matt trains for a total of 20 weeks.

d. How many kilometres will he have run in total in his entire training period?

Hint

Real-life situations can be modelled with series.
Any situation where a sequence of values increases or decreases by a constant amount can be modelled as an arithmetic sequence.

2. At the end of year 1, the adult population of a town is 28 000.
A model predicts that the adult population will increase by 2.5% each year.

a. Show that the predicted adult population at the end of year 2 is 28 700.

b. Write down an expression for the predicted population at the end of year n.

c. Find the predicted population at the end of year 10.

d. Give one criticism of this model.

Hint

Any real-life situation where a sequence of values increases or decreases in the same ratio can be modelled as a geometric sequence.
Numerical answers must be given to an appropriate degree of accuracy.
Answers to questions about modelling situations must be given in the context of the question.

3. E/P A girl saves money over a period of 52 weeks. She saves 10p in week 1, 15p in week 2, 20p in week 3 and so on until week 52. Her savings can be modelled as an arithmetic sequence.

a. Find the amount she saves in week 52. (3 marks)

b. Calculate her total savings over the 52-week period. (3 marks)

4. E/P Emma wants to take out a £6000 loan. She wants to make 24 monthly repayments on the loan. She finds these two options.
Option 1: Payments increase by a fixed amount each month. First payment is £120 and final payment is £396.
Option 2: First payment of £110. Payments increase by 7% each month.
Identify the cheapest loan option for Emma and work out the difference in the amount of interest charged. (8 marks)

5. E/P A car was purchased for £15 000 on 1st January. On 1st January each following year, the value of the car is 75% of its value on 1st January in the previous year.

a. Show that the value of the car exactly 3 years after it was purchased is £6328 to the nearest pound. (1 mark)

b. Find the number of years it takes after purchase for the value of the car to fall below £1000. (3 marks)

6. E/P The cost of home insurance with an insurance company for the first year is £250.
Every year the cost increases by 5%.

a. Find the cost of home insurance for the fourth year, giving your answer to the nearest penny. (2 marks)

b. Find the total cost of home insurance for the first 10 years. (3 marks)

The insurance company offers a discount on home insurance to customers who also purchase car insurance.

10% discount in annual payment up to a maximum of £35

c. Find the total cost for home insurance for the first 10 years with this discount. Give your answer to the nearest pound. (4 marks)

7. E/P A ball is dropped from a height of 2 metres. After each bounce, the ball reaches a height which is $\frac{3}{4}$ of its previous height. The successive heights of the ball can be modelled as a geometric series.

a. After how many bounces does the ball first bounce to a height which is less than 10 cm? (7 marks)

b. How far has the ball travelled in total when it hits the ground for the 10th time? (4 marks)

c. Give one criticism of this model for large numbers of bounces. (1 mark)

Problem solving: Set A

Bronze

An arithmetic sequence has 16th term 25, and 25th term 47.5.

a. Find the first term of the sequence. (3 marks)

The sum of the first n terms of the sequence is 712.5.

b. Find the value of n. (4 marks)

Silver

An arithmetic series is defined as $p + 2p + 3p + \ldots + 200$ where p is a constant.
Given that the sum of this series is 1700, find the value of p. (5 marks)

Gold

An arithmetic sequence has first three terms $3k + 2$, $5k + 7$ and $7k + 12$, where k is a constant.
Given that the 60th term of the sequence is positive, find the range of possible values for k. (5 marks)

Problem solving: Set B

Bronze

Find $\sum_{r=8}^{\infty} 3 \times \left(\frac{1}{2}\right)^r$, giving your answer as a fraction in simplest form. (5 marks)

Silver

Given that $\sum_{r=8}^{k}(2r - 5) = 299$, where k is a positive constant, find the value of k. (6 marks)

Gold

Given that $\sum_{r=1}^{16}(3r + 2^r + k) = 131\,550$, where k is a constant, find the value of k. (6 marks)

Now try this ⟶ Exam question bank Q24, Q27, Q61, Q64, Q68, Q73, Q76, Q80

Selected exam question bank questions

Question 24

The first term of an arithmetic series is 2. The common difference of the series is 6.

a. Find the 20th term of the series. (2 marks)

The sum of the first n terms of the series is 7450.

b. Find the value of n. (5 marks)

Question 27

A sequence $a_1, a_2, a_3, \ldots$ is defined by $a_1 = k$, $a_{n+1} = 3a_n - 5$, $n \geq 1$, where k is a constant.

a. Write down an expression for a_2 in terms of k. (1 mark)

b. Show that $a_3 = 9k - 20$. (2 marks)

Given that $\sum_{r=1}^{4} a_r = -30$,

c. find the value of k. (4 marks)

Question 61

a. Show that $\sum_{r=1}^{12}(2 + 4r + 3^r) = 797496$. (4 marks)

b. A sequence $u_1, u_2, u_3, \ldots$, is defined by $u_{n+1} = -\frac{1}{u_n}$, $u_1 = \frac{3}{4}$

Find the exact value of $\sum_{r=1}^{200} u_r$. (3 marks)

Question 64

The first term of a geometric series is 30 and the common ratio is $\frac{5}{8}$

a. Find, to one decimal place, the value of S_{15}. (2 marks)

The sum to infinity of the series is S_∞ and the sum to N terms of the series is S_N.

b. Find the smallest value of N, for which $S_\infty - S_N < 0.5$ (6 marks)

Question 68

A geometric series has first term 10 and common ratio $\frac{3}{5}$

Calculate:

a. the 20th term of the series, to 3 significant figures (2 marks)

b. the sum to infinity of the series. (2 marks)

Given that the sum to k terms of the series is greater than 24.99,

c. show that $k > \dfrac{\log 0.0004}{\log 0.6}$ (4 marks)

d. find the smallest possible value of k. (1 mark)

Question 73

A company made a profit of £60000 in 2018 (year 1). Future year-by-year profits are modelled by a geometric sequence with common ratio r, where $r > 1$.

The model predicts that the profits in 2019 (year 2) will be £60000r.

a. Write down an expression for the predicted profit in year n. (1 mark)

Given that the model predicts that the profit will exceed £240000 in year n,

b. show that $n > \dfrac{\log 4}{\log r} + 1$. (3 marks)

Using the model with $r = 1.08$,

c. find the year in which the profit made will first exceed £240000 (2 marks)

d. find the total of the profits that will be made by the company over the 10 years from 2018 to 2027 inclusive, giving your answer to the nearest £10000. (3 marks)

Question 76

The second and fourth terms of a geometric series are 270 and 120 respectively.

For this series, find:

a. the sum to infinity (6 marks)

b. the smallest value of n for which the sum of the first n terms of the series exceeds 1200. (4 marks)

Question 80

The first three terms of a geometric sequence are $6k - 5$, $4k - 5$ and $5 - k$, where k is a constant.

Given that $k < 1$,

a. show that $k = \dfrac{10}{11}$ (6 marks)

For this value of k,

b.

i. evaluate the fourth term of the sequence, giving your answer as an exact fraction

ii. evaluate the sum of the first ten terms of the sequence.

(6 marks)

Answers

3.1 Arithmetic sequences

1.

a.

i. $7, 11, 15, 19$

ii. $a = 7, d = 4$

b.

i. $4, 1, -2, -5$

ii. $a = 4, d = -3$

c.

i. $8.5, 9, 9.5, 10$

ii. $a = 8.5, d = 0.5$

d.

i. $-3, -1, 1, 3$

ii. $a = -3, d = 2$

2.

a. nth term $= 4n - 2$, 10th term $= 38$

b. nth term $= 7 - 2n$, 10th term $= -13$

c. nth term $= 3n - 5$, 10th term $= 25$

d. nth term $= 11y - 3ny$, 10th term $= -19y$

3.

a. 33

b. 41

c. 30

d. 53

4. $a = 2, d = 5$

5.

a. 8

b. 41

6.

a. $-1, 2, 5$

b. 3

7.

a. $a = 10, d = -4$

b. 17

8.

a. Let the terms be $a = \alpha + kd$, $b = \alpha + (k+1)d$ and $c = \alpha + (k+2)d$

$$\frac{a+c}{2} = \frac{\alpha + kd + \alpha + (k+2)d}{2} = \frac{2\alpha + 2(k+1)d}{2} = \alpha + (k+1)d = b$$

so $\dfrac{a+c}{2} = b$

b. $p = -3, p = 2.5$

9.

a. $k = 4$

b. 445

3.2 Arithmetic series

1.

a. 860

b. 570

c. -1080

d. 50

2.

a. 456

b. -816

c. 5190

d. 960

3. $2S_{60} = 60 \times 61$, so $S_{60} = \dfrac{60 \times 61}{2} = 1830$

4. 25

5.

a. $-1, 3, 7$

b. 4

c. $S_n = \dfrac{n}{2}(a + l)$ where $a = -1, d = 4$ and $l = 4n - 5$
So $S_n = \dfrac{n}{2}(-1 + 4n - 5) = \dfrac{n}{2}(4n - 6)$
which shows $S_n = n(2n - 3)$

6.

a. $S_n = 1 + 2 + 3 + \cdots + (n-2) + (n-1) + n$
$S_n = n + (n-1) + (n-2) + \cdots + 3 + 2 + 1$
$2S_n = n \times (n+1)$ so $S_n = \dfrac{n(n+1)}{2}$

b. $S = \dfrac{100 \times 202}{2} = 10100$

7.

a. 40

b. 50

8.

a. $S_{12} = \dfrac{12}{2}(2a + (12-1)d) = 366$
So $6(2a + 11d) = 366$, which gives $12a + 66d = 366$

b. $a + 7d = 38$

c. $a = 3, d = 5$

9.

a. Second term $= a + d$, sixth term $= a + 5d$,
so $a + d + a + 5d = 2a + 6d = 14$
so $a + 3d = 7$

b. $a = 13, d = -2$

10.

a. $a + 14d = 11$ (1), $a + 19d = \dfrac{37}{2}$ (2)
subtracting, (2) $-$ (1) gives $5d = \dfrac{15}{2}$, so $d = \dfrac{3}{2}$
substituting into (1) gives $a + 14 \times \dfrac{3}{2} = 11$
$a = 11 - 21 = -10$

Second term $= -10 + \frac{3}{2} = -\frac{17}{2}$

b. $n = 60$

11. $d = k + 5 = 3k - 5$ so $2k = 10$ and $k = 5$.
Terms are 15, 25 and 35 so $d = 10$.
First term $= 15 - 10 = 5$
$S_n = \frac{n}{2}(10 - 10(n-1))$
$= \frac{10n}{2}(1 + (n-1)) = 5n \times n = 5n^2$

12. $d = 4k + 3 - 3k = 6k - 9 - (4k + 3)$,
so $k + 3 = 2k - 12$ and $k = 15$.
$d = k + 3 = 18$
Third term $= 3k = 45 = a + 2d$ so $a = 45 - 36 = 9$
$S_n = \frac{n}{2}(2a + (n-1)d)$
$= S_n = \frac{n}{2}(18 + (n-1) \times 18) = \frac{n}{2} \times 18n = 9n^2 = (3n)^2$
So the sum of the first n terms of the sequence is a square number.

3.3 Geometric sequences

1.

a.

i. $r = 3$

ii. $u_n = 2 \times 3^{n-1}$

b.

i. $r = \frac{1}{2}$

ii. $u_n = 60 \times \left(\frac{1}{2}\right)^{n-1}$

c.

i. $r = -2$

ii. $u_n = 10 \times (-2)^{n-1}$

d.

i. $r = \frac{3}{2}$

ii. $u_n = 40 \times \left(\frac{3}{2}\right)^{n-1}$

2.

a. Part c: $10, -20, 40, -80$

b. Let the terms be $p = ar^k, q = ar^{k+1}, r = ar^{k+2}$, then
$pr = ar^k \times ar^{k+2} = a^2r^{2k+2} = (ar^{k+1})^2 = q^2$, so $pr = q^2$

3.

a. $x = 10$

b. 2560

4. 2125764 (13th term)

5.

a. $\frac{5}{2}$

b. $\frac{128}{125}$

6. $k = \frac{1}{2}$

7.

a. 0.110 (3 s.f.)

b. 0.832 (3 s.f.) (15th term)

8.

a. $a = 150, ar^4 = 50$ so $150r^4 = 50$ which gives $r^4 = \frac{1}{3}$
Taking logs on both sides, $4\ln r = -\ln 3$
so $4\ln r + \ln 3 = 0$

b. 0.760

9.

a. $k = \frac{1}{2}$

b. 10240 (13th term)

3.4 Geometric series

1.

a. 59048

b. 96.0 (3 s.f.)

c. -255

d. 2370 (3 s.f.)

2.

a. 1275

b. 19682

c. 819 (3 s.f.)

d. 4118

3. 19

4.

a. 2

b. 3

c. 3145725

5.

a. 0.101

b. 4

6.

a. $S_k = \frac{20\left(1 - \left(\frac{4}{5}\right)^k\right)}{1 - \frac{4}{5}}$
So $100\left(1 - \left(\frac{4}{5}\right)^k\right) > 50 \Rightarrow 1 - \left(\frac{4}{5}\right)^k > 0.5$
$\Rightarrow 0.5 > \left(\frac{4}{5}\right)^k$
Taking logs gives $\ln 0.5 > k \ln \frac{4}{5}$, so $k > \frac{\ln 0.5}{\ln 0.8}$

b. $k = 4$

7.

a. $k = 10$

b. 0.0677

c. 63.8

8. 536 (3 s.f.)

9.

a. $a + ar = ar^2$ where a is the first term and r is the common ratio. So $r^2 - r - 1 = 0$, hence r is independent of a.

b. $S_{20} = 48900$ (3 s.f.)

3.5 Sum to infinity

1.

a. $r = 0.2 \div 2 = 0.1$
As $|r| < 1$, this means the series is convergent.

b. $S_\infty = \dfrac{20}{9}$

2.

a.

i. $|r| = 0.4 < 1$, so the series is convergent.

ii. $S_\infty = \dfrac{25}{3}$

b.

i. Divergent, since $|r| = 3 > 1$

c.

i. Convergent, since $|r| = \dfrac{1}{3} < 1$

ii. $S_\infty = \dfrac{243}{4}$

d.

i. Not convergent: series is arithmetic.

3.

a. $ar^4 = 2.4576, ar^6 = 1.572864$
$\dfrac{ar^6}{ar^4} = \dfrac{1.572864}{2.4576}$, so $r^2 = 0.64$ and $r = \pm 0.8$
As $|r| < 1$ this means the series is convergent.

b. 30 or $\dfrac{10}{3}$

4.

a. 0.68

b. 32

5.

a. $r = \pm\dfrac{1}{3}$

b. $a = 54$

6. $\dfrac{2}{3}$

7.

a. $|r| < 1$

b. $\dfrac{3}{4}$

8.

a. 0.9873

b. 3.8

9.

a. $\dfrac{1}{3}$

b. 1.98

c. 119.8 (1 d.p.)

10.

a. $ar = \frac{12}{5}$; $\frac{a}{1-r} = 10$

So $\frac{12}{5} = 10r(1-r)$, which gives $50r^2 - 50r + 12 = 0$

b. $r = \frac{2}{5}$ or $r = \frac{3}{5}$

c. $a = 6$ or $a = 4$

d. $n = 8$

3.6 Sigma notation

1.

a.

i. $1, 5, 9, 13, 17, 21$

ii. 66

b.

i. $2, 16, 54, 128, 250$

ii. 450

c.

i. $1, 0, -1, 0, 1$

ii. 1

d.

i. $-\frac{3}{8}, \frac{3}{16}, -\frac{3}{32}, \frac{3}{64}, -\frac{3}{128}$

ii. $-\frac{33}{128}$

2.

a. $a = 4, d = 5$

b. 378

3.

a. $a = 3, r = 2$

b. 3145725

4.

a. 185

b. 60

c. 10

5.

a. $\sum_{r=1}^{30}(5-2r) = \sum_{r=1}^{7}(5-2r) + \sum_{r=8}^{30}(5-2r)$

so $\sum_{r=8}^{30}(5-2r) = \sum_{r=1}^{30}(5-2r) - \sum_{r=1}^{7}(5-2r)$

b. $\sum_{r=1}^{30}(5-2r)$: $a = 3, d = -2, n = 30$,

$S_{30} = \frac{30}{2}(6 + 29 \times (-2)) = -780$

$\sum_{r=1}^{7}(5-2r)$: $a=3, d=-2, n=7$,
$S_7 = \frac{7}{2}(6+6\times(-2)) = -21$
$\sum_{r=8}^{30}(5-2r) = -780-(-21) = -759$

6. 28658448

7. 885720

8.

a. $a=9, d=3, S_k = \frac{k}{2}(18+(k-1)\times 3) = 750$,
so $k(18+3k-3) = 1500$, which gives
$3k^2+15k-1500 = 0$
$\Rightarrow (k-20)(3k+75) = 0$

b. $k = 20$

9.

a. $a=8, r=2$,
$S_k = \frac{8(2^k-1)}{2-1} = 8(2^k-1) = 262136$
So $(2^k-1) = 32767$ and $2^k = 32768$
Taking logs gives $k\ln 2 = \ln 32768$,
$\Rightarrow k = \frac{\ln 32768}{\ln 2}$

b. $\sum_{r=1}^{15} 4\times 2^r = 262136$
So $\sum_{r=16}^{20} 4\times 2^r = \sum_{r=1}^{20} 4\times 2^r - \sum_{r=1}^{15} 4\times 2^r$
$= 8388600 - 262136 = 8126464$

10.

a. $|x| < \frac{1}{4}$

b. $\frac{1}{5}$

11. $\sum_{r=1}^{10} 4 = 4\times 10 = 40$
$\sum_{r=1}^{10} 3r = 3\sum_{r=1}^{10} r = 3\times\frac{10\times 11}{2} = 165$
$\sum_{r=1}^{10} 2^{r-1} = \frac{1(2^{10}-1)}{2-1} = 1023$
$40+165+1023 = 1228$

3.7 Recurrence relations

1.

a. $3, 5, 9, 17$

b. $10, 6, 2, -2$

c. $4, 8, 32, 512$

2.

a.

i. increasing

b.

i. periodic

ii. order 4

c.

i. decreasing

3.

a.

i. $2, 5, 8, 11, 14$

ii. increasing

b.

i. $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$

ii. decreasing

c.

i. $\frac{1}{2}, -\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, \frac{1}{2}$

ii. periodic

iii. 2

d.

i. $1, 3, 7, 15, 31$

ii. increasing

e.

i. $8, -4, 8, -4, 8$

ii. periodic

iii. 2

f.

i. $10, 6, 2, -2, -6$

ii. decreasing

4.

a. $n = 1$: $a_2 = k + (-1)^1 = k - 1$
$n = 2$: $a_3 = k - 1 + (-1)^2 = k - 1 + 1 = k$
$n = 3$: $a_4 = k + (-1)^3 = k - 1$, so $a_2 = a_4 = k - 1$

b. 1000

c. $a_{399} = 3$

5. $p = 2, q = -5$

6.

a. $u_1 = 5, u_2 = 5k - 8, u_3 = k(5k - 8) - 8 = 5k^2 - 8k - 8$

b. $k = 4$

c. 600

7.

a. 4

b. 0

8.

a. $a_3 = 25k + 24$

b. $a_4 = 5(25k + 24) + 4 = 125k + 124$
$\sum_{r=1}^{4} a_r = k + 5k + 4 + 25k + 24 + 125k + 124$

$= 156k + 152 = 4(39k + 38)$, so a multiple of 4

9.

a. 75

b. 150

3.8 Modelling with series

1.

a. Sundays: first = 6 km, second = $6 + 2 = 8$ km,
third = $8 + 2 = 10$ km, fourth = $10 + 2 = 12$ km,
fifth = $12 + 2 = 14$ km

b. $2n + 4$

c. 19 weeks

d. 500 km

2.

a. $1.025 \times 28000 = 28700$

b. $28000 \times 1.025^{n-1}$

c. 34968 (nearest whole number)

d. It is unlikely that the adult population will increase by exactly the same percentage each year.

3.

a. £2.65

b. £71.50

4. Option 1 is cheaper. Difference in interest is £207.

5.

a. $15000 \times 0.75^3 = 6328.125 =$ £6328 to the nearest pound

b. 10

6.

a. £289.41

b. £3144.47 to the nearest penny

c. £2836

7.

a. 11

b. 13.10 m to the nearest cm

c. The ball is unlikely to continue to bounce to a height which is $\frac{3}{4}$ of its previous height for a long period of time due to energy losses.

Problem solving: Set A

Bronze a -12.5 b $n = 30$

Silver 12.5

Gold $k > -\frac{27}{11}$

Problem solving: Set B

Bronze $\frac{3}{128}$

Silver $k = 20$

Gold $k = \dfrac{9}{2}$

Selected exam question bank answers

Question 24

a. 116

b. $S_n = \dfrac{n}{2}(4 + (n-1) \times 6) = 7450,$
So $n(6n - 2) = 14900$
$\Rightarrow 6n^2 - 2n - 14900 = 0$
which gives $3n^2 - n - 7450 = 0$
So $n = 50$

Question 27

a. $a_2 = 3k - 5$

b. $a_3 = 3(3k - 5) - 5 = 9k - 15 - 5 = 9k - 20$

c. $k = 1.5$

Question 61

a. $\displaystyle\sum_{r=1}^{12} 2 + \sum_{r=1}^{12} 4r + \sum_{r=1}^{12} 3^r$
$= (2 \times 12) + \dfrac{12}{2}(8 + 11 \times 4) + \dfrac{3(3^{12} - 1)}{3 - 1}$
$= 797496$

b. $u_1 = \dfrac{3}{4}, u_2 = -\dfrac{4}{3}, u_3 = \dfrac{3}{4}, u_5 = \dfrac{4}{3} \ldots$
$\displaystyle\sum_{r=1}^{200} u_r = 100 \times \left(\frac{3}{4} - \frac{4}{3}\right) = -\frac{175}{3}$

Question 64

a. 79.9

b. $N = 11$

Question 68

a. 0.000609

b. 25

c. $S_k = \dfrac{10(1 - 0.6^k)}{(1 - 0.6)} = 25(1 - 0.6^k) > 24.99$
$0.01 > 25 \times 0.6^k$
Dividing both sides by 25 gives $0.0004 > 0.6^k$
So $\log 0.0004 > k \log 0.6$
$\Rightarrow \dfrac{\log 0.0004}{\log 0.6} < k$
(direction of inequality changed since $\log 0.6$ is negative)
So $k > \dfrac{\log 0.0004}{\log 0.6}$

d. 16

Question 73

a. $60000r^{n-1}$

b. $60000r^{n-1} > 240000$, so $r^{n-1} > 4$,
Taking logs: $(n-1)\log r > \log 4$
$n - 1 > \dfrac{\log 4}{\log r}$
$n > \dfrac{\log 4}{\log r} + 1$

c. Year 20

d. £870000

Question 76

a. 1215

b. 11

Question 80

a. $(4k-5)^2 = (6k-5)(5-k)$
so $16k^2 - 40k + 25 = 30k - 6k^2 - 25 + 5k$
so $22k^2 - 75k + 50 = 0$
$(11k-10)(2k-5) = 0$ so $k = \dfrac{5}{2}$ or $\dfrac{10}{11}$
As $k < 1$, $k = \dfrac{10}{11}$

b.

i. $-\dfrac{135}{11}$

ii. -6710